VIGNAN'S INSTITUTE OF INFORMATION TECHNOLOGY (AUTONOMOUS)

HACKMATRIX 1.0

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE DEPARTMENT

PS-01 — Smart Transportation & Mobility

Description

Traffic congestion, long travel times, inefficient routes, overcrowded public transport, and unexpected delays are common transportation problems. These problems can change depending on the **time, location, traffic conditions, and travel demand**.

The challenge is to develop a **smart transportation solution** that can analyze these conditions and help people or transportation authorities make better decisions about **routes, travel, and mobility**.

Objectives

Participants are expected to develop a solution that can:

- Identify traffic congestion and transportation bottlenecks.
- Analyze travel and traffic patterns.
- Suggest suitable and efficient routes.
- Identify or predict possible travel delays.
- Suggest alternative routes or transportation options.
- Improve the use of available transportation resources.
- Support better transportation planning and decision-making.

PS-02 — AI-Based Energy Consumption Optimizer

Description

Buildings, industries, and other facilities use energy for lighting, cooling, heating, machinery, and electrical equipment. Energy consumption can vary depending on **time, usage patterns, occupancy, weather, and operating conditions**.

Unnecessary or unusually high energy consumption can increase costs and reduce energy efficiency. The challenge is to develop an **AI-based energy consumption solution** that can understand energy usage patterns, identify inefficient consumption, and help users take suitable actions to reduce energy usage.

Objectives

Participants are expected to develop a solution that can:

- Analyze energy consumption patterns.
- Identify unusually high or inefficient energy usage.
- Identify periods of high energy demand.
- Predict future energy consumption where applicable.
- Identify possible reasons for excessive energy usage.
- Recommend suitable energy-saving actions.
- Estimate the possible energy or cost savings from recommended actions.

PS-03 — GenAI Resume & Job Matching Assistant

Description

Job seekers often find it difficult to understand whether their **skills, education, experience, and qualifications** match the requirements of a particular job. Recruiters also spend considerable time reviewing resumes and identifying candidates who are suitable for specific roles.

The challenge is to develop a **Generative AI-powered Resume and Job Matching Assistant** that can understand a candidate's resume and a job description, compare their requirements, and clearly show how well the candidate matches the role.

Objectives

Participants are expected to develop a solution that can:

- Understand important information from a resume.
- Understand the requirements mentioned in a job description.
- Identify matching skills, qualifications, and experience.
- Identify missing or insufficient skills.
- Provide an overall job-match assessment.
- Explain the reasons behind the assessment.
- Suggest areas where the candidate can improve their profile.

PS-04 — AI Agent for Repairing Broken APIs

Description

When an API stops working, developers usually need to check **error messages, logs, and source code** to find the cause of the problem. After identifying the issue, they need to make the required code changes and test whether the problem has been fixed. This process can take significant time, especially when the cause of the failure is not immediately clear.

The challenge is to develop an **AI-powered agent that can detect API failures, investigate the possible cause, suggest or generate a code fix, and verify the fix through testing**.

Objectives

Participants are expected to develop a solution that can:

- Detect API failures or errors.
- Analyze logs, error messages, and relevant application information.
- Identify the possible cause of an API failure.
- Locate the code related to the problem.
- Suggest or generate an appropriate code fix.
- Test the proposed fix automatically.
- Present the identified problem and proposed solution clearly for developer review.

PS-05 — AI-Powered Plant Disease & Crop Health Assistant

Description

Farmers may find it difficult to identify crop diseases and unhealthy plants at an early stage. When a problem is not identified quickly, it can spread and affect **crop quality, productivity, and overall yield**.

The challenge is to develop an **AI-powered crop health solution** that can analyze plant-related information and help identify whether a crop is healthy or showing signs of disease or other health problems.

Objectives

Participants are expected to develop a solution that can:

- Identify whether a plant is healthy or unhealthy.
- Detect possible diseases or visible crop problems.
- Classify different crop diseases or health conditions.
- Analyze plant images or other relevant information.
- Provide a clear explanation of the identified condition.
- Suggest suitable next steps or preventive measures.

PS-06 — Smart Complaint Management System

Description

Students regularly face issues related to Wi-Fi, classrooms, laboratories, hostels, transportation, washrooms, electrical systems, and other campus facilities. Managing these complaints manually can make it difficult to identify the right department, prioritize urgent issues, track progress, and identify recurring problems.

The challenge is to develop a **Smart Complaint Management System** that can understand student complaints, organize them appropriately, and help ensure that complaints are efficiently handled and resolved.

Objectives

Participants are expected to develop a solution that can:

- Allow students to report campus-related complaints easily.
- Identify and categorize complaints based on their nature.
- Determine the priority or urgency of a complaint.
- Identify the appropriate department or authority responsible for handling it.
- Identify similar or duplicate complaints.
- Allow users to track the status of their complaints.
- Provide useful insights into recurring complaints and campus issues.

PS-07 — Smart Lost and Found Management System

Description

Students, faculty, and staff frequently lose or find belongings such as ID cards, books, wallets, bags, keys, electronic devices, certificates, and other personal items within the campus. Identifying the owner of a found item or finding a lost item can be difficult when information is scattered or manually managed.

The challenge is to develop a **Smart Lost and Found Management System** that helps users report lost or found items and intelligently identify possible matches between them.

Objectives

Participants are expected to develop a solution that can:

- Allow users to report lost and found items.
- Capture relevant information about an item.
- Identify similarities between lost and found item reports.
- Suggest possible matches between reported items.
- Help users verify potential matches.
- Track the status of lost and found cases.
- Provide a centralized platform for managing lost and found items.

PS-08 — Intelligent Faculty Workload Balancing System

Description

Faculty members are assigned different academic and institutional responsibilities, including teaching, laboratory sessions, project guidance, examinations, mentoring, and departmental activities. Manual allocation of these responsibilities can result in an uneven workload, where some faculty members are overloaded while others are underutilized.

The challenge is to develop an **Intelligent Faculty Workload Balancing System** that analyzes assigned responsibilities and helps distribute the workload appropriately according to faculty designation and applicable workload norms.

For this challenge, the prescribed workload limits are:

- Professor — 10 hours
- Associate Professor — 16 hours
- Assistant Professor — 20 hours

Objectives

Participants are expected to develop a solution that can:

- Calculate the workload assigned to each faculty member.
- Consider workload limits based on faculty designation.
- Identify overloaded, balanced, and underloaded faculty members.
- Consider different types of academic and institutional responsibilities.
- Recommend a more balanced distribution of tasks.
- Maintain academic and departmental requirements while reallocating responsibilities.